Experiment 1: BASIC LABORATORY TECHNIQUES AND OPERATIONS

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Abstract

This experiment was made and took part to familiarize students with fundamental laboratory techniques, including the accurate measurement of mass, length, volume, density, and temperature. The hypothesis tested is that utilizing specialized quantitative laboratory instruments (such as graduated cylinders and non-luminous flames) yields significantly higher precision and accuracy than using basic or non-standard tools. The methodology involved measuring the mass of one-peso coins and sodium chloride using a balance, calculating the area of a laboratory manual, determining the density of solid and liquid samples via geometric and displacement methods, and analyzing the heating efficiency of luminous versus non-luminous Bunsen burner flames. Results supported the hypothesis, indicating variations in precision depending on the instruments used; for instance, using a spoon instead of a laboratory scoop yielded less precise mass measurements for sodium chloride. Furthermore, the non-luminous flame proved significantly more efficient for heating, boiling 5 mL of water in 13.86 seconds compared to 21.89 seconds for the luminous flame. Several measurement and calculation discrepancies were identified, highlighting the importance of proper unit recording and formula application. Ultimately, the experiment successfully reinforced essential technical skills required for accurate data collection and laboratory safety.

Keywords: Measurement, Density, Bunsen Burner, Precision, Laboratory Techniques

1. Introduction

Accuracy and precision in laboratory measurements are the foundational elements of experimental science. A thorough understanding of how to correctly read scientific instruments and minimize human error is necessary to produce reliable data. This study addresses the operational differences between various measuring devices and heating methods, such as comparing the efficacy of luminous and non-luminous flames. By systematically performing basic measurements and calculations, this experiment attempts to test the precision of standard laboratory glassware and highlight common sources of human and instrumental error in data gathering. The overall goal of this experiment is to develop technical proficiency in basic laboratory operations, specifically regarding the measurement of mass, length, volume, density, and temperature.

2. Methodology

2.1. Mass Measurement

Five (5) one-peso coins were obtained, then the laboratory balance was properly checked beforehand to ensure that it was calibrated or zeroed before use. The mass of each coin was measured individually and recorded, then the five coins were measured together to determine their combined mass. A clean and dry watch glass was then weighed and its mass was recorded. Using a clean spatula or scoop, one heaping scoop of sodium chloride (NaCl) was placed on the watch glass, and the combined mass was measured. A second heaping scoop of NaCl was added and the combined mass was recorded once again, followed by a third scoop and a final mass measurement. After all the measurements were accomplished, the sodium chloride was properly disposed in a designated container for used NaCl. The watch glass was cleaned, and all the materials used were returned to the laboratory tray we used.

2.2. Length Measurement

A metric ruler was used to measure the following: Total length of the test tube, the length and width of the laboratory manual, and the height of a 100-ml graduated cylinder including its base. All dimensions were measured and recorded in both centimeters (cm) and inches (in) then recorded to the nearest graduation indicated by the ruler, and was reported using the appropriate amount of significant figures. The glassware was carefully handled during measurement to prevent accidental breakage.

2.3. Volume Measurement

The capacity of the selected laboratory glassware was measured using distilled water. A clean test tube was filled completely with distilled water, then the water was transferred into a graduated cylinder to determine and record its volume. The same procedure was repeated using a 100-mL beaker instead of a test tube, and the measured capacities of the test tube and beaker were compared

In verifying the calibration of the beaker, distilled water was poured into a 150-mL beaker until the liquid reached the 50 mL graduation mark. The water was then transferred into a graduated cylinder, and the actual volume was measured and recorded. The indicated volume on the beaker was then compared with the volume measured using the graduated cylinder.

2.4. Density Measurement

The density measurement was separated into two (2) parts: Density of a Solid and Density of Water. For the density of a solid, two methods were used: (a.) geometric measurement and (b.) water displacement method.

In geometric measurement, ten (10) one-peso coins were stacked on top of each other neatly, then the combined mass of the coins was measured and recorded. The diameter and total height of the stack of coins were measured, while the radius was calculated from the measured diameter. The volume of the cylindrical stack was determined using $V = \pi r^2 h$ and the density using $D = m/V$.

For the water displacement method, a graduated cylinder was filled up with approximately 50 mL of distilled water, then the initial volume was recorded. The ten (10) one-peso coins were then carefully transferred into the graduated cylinder without splashing the water, after which the final volume was recorded. The volume of the coins was determined from subtracting the initial volume from the final volume, and the density of the coins was calculated using the measured mass and displaced volume.

To determine the density of water, the mass of a clean and dry 150-mL beaker was measured and recorded. Exactly 50.0 mL of distilled water was measured using a graduated cylinder and was transferred into the beaker. The combined mass of the beaker and water was then measured. The mass of the water was determined through subtracting the mass of the empty beaker from the combined mass, and the density of water at room temperature was calculated using the measured mass and volume.

2.5. Bunsen Burner Operation

The parts of the bunsen burner were identified, including the barrel, collar, air holes, gas inlet, and base. The burner was assembled and connected securely to the gas supply using rubber tubing. Before ignition, the air holes were closed, and the burner was lit by placing a lighted match or striker near the top of the barrel while the gas valve was slowly opened. The characteristics of the luminous flame were then observed. Approximately 5 mL of distilled water was placed in a test tube and heated using the luminous flame, while the time required for the water to boil were observed and recorded. The air holes were gradually opened to produce a non-luminous flame, and the heating procedure was repeated using another 5 mL sample of distilled water in a test tube. The boiling time of the water in luminous and non-luminous flame were recorded and compared. Finally, a clean iron nail was held with crucible tongs and heated first in the outer cone and then in the inner cone of the non-luminous flame, and the observed changes were compared.

2.6. Temperature Measurement

A ring stand was assembled with with an iron ring and wire gauze, and a 150-mL beaker was filled approximately halfway with distilled water and placed on the wire gauze. The water was heated using a properly adjusted non-luminous flame. A thermometer was suspended so that the sensing bulb was completely submerged in the water without touching the bottom or sides of the beaker. The boiling temperature of the water was recorded in degrees Celsius (℃), after which the recorded temperature was converted to degrees Fahrenheit (℉). After all the measurements were completed, the gas supply was turned off, and the apparatus was left to cool off before the setup was disassembled.

3. Results and Discussion

The results obtained from the different laboratory activities are presented below. The measurements included mass, length, volume, density, Bunsen burner heating time, and temperature.

3.1 **Mass Measurement**

For the one-peso coins, the individual masses recorded were 6.79 g, 6.29 g, 6.09 g, 6.19 g, and 5.49 g, respectively. The total mass of the individual coins was 30.9 g, while the five coins weighed together had a measured mass of 27.7 g. The calculated average mass of one coin was 6.18 g.

The sodium chloride measurements were obtained using successive scoops. The calculated masses of the first, second, and third scoops were 2.9 g, 5.4 g, and 8.1 g, respectively. The average mass of one heaping scoop was recorded as 5.47 g.

The electronic balance demonstrated high repeatability, through a minor 3.2 g variance occurred between the cumulative mass of individual coins (30.9g) and their group measurement (27.7 g), likely due to surface residue or balance calibration drift. For sodium chloride handling, successive scoops yielded consistent mass increments (averaging 2.7 g), proving that standardized sampling tools like spatulas provide uniform sampling. For length measurements, precision was bounded by the 0.1 cm ruler graduation. Using the correct conversion factor (1 in. = 2.54 cm) yields accurate dimensions for the test tube (5.71 in.) and lab manual (10.98 in. x 8.50 in.), preserving significant figures.

3.2 **Length Measurement**

The measured dimensions of the laboratory materials are shown below.

Object	Length (cm)	Length (in.)
Test tube	**14.5**	**57.1**
Graduated cylinder	**24.3**	**96.70**
Laboratory manual – Length	**27.9**	**109.80**
Laboratory manual – Width	**21.6**	**85.00**

Using the recorded length and width of the laboratory manual, the calculated area was 602.64 cm^2.

Comparing glassware capacities highlights the distinction between containment vessels and quantitative instruments. Pouring water to the 50-mL mark on a 150-mL beaker delivered only 49 mL into a graduated cylinder. Beakers have wide cross-sectional diameters with coarse graduation marks prone to visual parallax error. Graduated cylinders feature narrow bores and finer calibrations, making them necessary for precise liquid measurement.

3.3 **Volume Measurement**

The measured volumes of the different glassware were recorded as follows:

Glassware	Measured Volume (mL)
Test tube	**29 mL**
100-mL beaker	**99 mL**
150-mL beaker	**148 mL**
Graduated cylinder verification	**approximately 49 mL**

The geometric density calculation for the coin stack (0.083 g/ cm^3) was heavily distorted by a recorded stack height error (15 cm instead of 1.5 cm) and unmeasured air gaps between worn surfaces. Correcting the height yields a realistic density of = 7.2 g/cm^3. The water displacement method yielded 9.1 g/cm^3 (54.6 g/ 6 mL), effectively bypassing irregular geometries despite 1-mL meniscus reading limits. For liquid density, the measured water mass (34.5 g) over volume (34.5 mL) gave 1.00 g/mL, matching pure water's theoretical density (= 0.998 g/mL).

3.4 **Density Measurement**

For the geometric method, the measured mass of the coin stack was 54.6 g, with a recorded diameter of 23.10 cm and a height of 15 cm. The calculated volume was recorded as 629.44 cm^3, resulting in a calculated density of approximately 8.3×10^{-2} g/cm^3.

For the water displacement method, the initial water volume was 50 mL, while the final volume was 56 mL. The resulting displaced volume of the coins was 6 mL, and the recorded density was 9 g/cm^3.

For the density of water, the mass of the empty beaker was 65.5 g, while the mass of the beaker containing water was 100.0 g. The recorded volume of water was 34.5 mL, giving a calculated density of approximately 0.00606 g/cm^3.

Closing the air collar restricts oxygen intake, resulting in incomplete hydrocarbon combustion (CH4 + O2 (arrow) C (s) + CO (g) + 2H2O). This produced a cooler, soot-depositing luminous flame that took 21.89 seconds to boil 5 mL of water. Opening the air collar provided complete oxidation (CH4 + 2O2 (arrow) CO2 + 2H2O), creating a hotter, non-luminous blue flame that boiled the sample in 13.86 seconds without soot. Heating an iron nail confirmed the inner cone tip as the hottest zone.

3.5 **Bunsen Burner Operation**

The luminous flame was observed to have a yellow/orange color and produced soot or smoke. The non-luminous flame was observed to have a blue color and appeared cleaner.

The heating test showed that 5 mL of water required 21.89 seconds to boil using the luminous flame, while the same volume required 13.86 seconds using the non-luminous flame.

The observed boiling point of water (98.0°C/208.4°F) deviated by 2.0°C from standard conditions (100.0°C), which is attributable to local atmospheric pressure fluctuations and thermometer calibration offsets. Key experimental errors across all trials included meniscus parallax error, ambient radiative thermal loss during heating, and initial unit recording mistakes.

3.6 **Temperature Measurement**

The recorded boiling point of water was 98 °C. After conversion, the corresponding temperature was 208.4 °F.

The observed boiling point of distilled water was recorded as 98.0°C (208.4°F). The slight 2.0°C deviation from the standard theoretical boiling point (100.0°C) is attributable to local atmospheric pressure variations and minor calibration offsets in the laboratory thermometer. Across all experimental trials, primary sources of error included parallax error during meniscus readings in wide-bore glassware, radiative thermal loss to ambient air during open-beaker heating, and initial recording/calculation mistakes.

4. Conclusion

- In conclusion, this experiment was able to effectively highlight the importance of accurate measurement and correct laboratory technique to getting accurate experimental results. The experiments involving mass, length, volume, density, temperature and Bunsen burner use were able to help contribute in learning the most correct way of using a given laboratory instrument and the importance of correct calculations and recording of unit of measurements. The non-luminous Bunsen burner flame proved to be the more efficient one for heating and boiling 5 mL of water, boiling the 5 mL of water in 13.86 seconds as compared to 21.89 seconds when using the luminous Bunsen burner. The resulting calculated density for water was around 1.00 g/mL and results from the coin experiments really demonstrated the concept that errors in measurements and recording can impact the calculated values quite significantly. The experiment was successfully able to meet the set learning objectives by making the readers learn and understand the use of the different laboratory instruments and the related calculations involved while emphasizing on the need to be precise in measurements and technique while keeping in mind the potential sources of errors.

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Criteria for Grading Lab Reports

Overall (OA): 5 points
Does it contain the required sections?
Is it clearly written?
Does it use good grammar and scientific terms properly?
Are the words spelled correctly?
Are the calculations performed correctly?
Does the title page contain the author's name and address?

Abstract (A): 5 points
Does it introduce the overall topic?
Does it explain the hypothesis being tested?
Are the important methods described?
Does it reach logical conclusions supported by the data?
Does it flow well? Is it logically written? Is it concise?

Introduction (I): 10 points
Does it give general background?
Does it point out poorly understood or unknown factors related to the study?
Does it raise questions?
Does it explain the hypothesis being tested? Does it contain the objectives of the experiment?
Does it flow well? Is it logically written? Is it concise?

Materials and Methods (M): 5 points
Could the experiments be understood based on the information given?
Does it include the source of the reagents?
Does it describe all of the methods used?
Is it excessively long?

Results and Discussion (R&D): 20 points
Does it explain the rationale and strategy for the experiments performed?
Does it describe, in words, what was done?
Does it summarize the findings obtained in the Results section?
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Does it discuss the expected and unexpected results?
Figures and Tables: Are the figures and tables well designed?
Do the figures include informative legends?
Do the figures present information useful for understanding the text?

Conclusion (C): 5 points
Does it answer the questions raised in the Introduction?
Does it answer the objectives contained in the Introduction?
Does it reach conclusions?
Does it explain why the conclusions are important?